FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

FULL TEST – III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 264**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Three sections: **Section-A Section-B & Section-C.**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section–A (01 – 10, 21 – 30, 41 – 50)** contains 30 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

2. **Section–B (11 – 12, 31 – 32, 51 – 52)** contains 6 Match the following Type questions. Each question having 4 statements in Column I & 5 statements in Column II with any given statement in Column I having correct matching with 1 or more statement (s) given in Column II. Each statement carries **+2 marks** for correct answer and **–1 mark** for wrong answer.

3. **Section–C (13 – 20, 33 – 40, 53 – 60)** contains 24 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

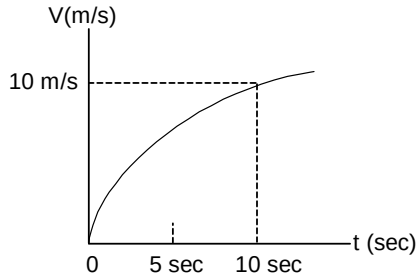
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **10** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A particle is moving in such a way that its velocity versus time graph is a parabola ($t = kv^2$, k is constant) as shown in the figure. Choose the correct option(s)



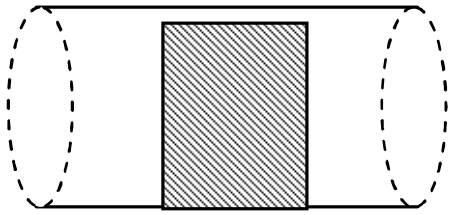
(A) acceleration of the particle is $\sqrt{3}\,m/s^2$ throughout the journey

(B) average acceleration of particle is 1 m/s^2 for the interval of first 5 sec.

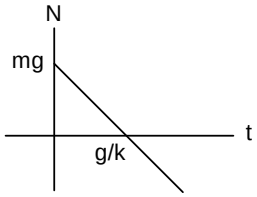
(C) average acceleration of particle is less than $\sqrt{3}\,m/s^2$ for the interval of first 5 sec.

(D) average acceleration of particle is less than 1 m/s^2 for the interval of 5 sec to 10 sec.

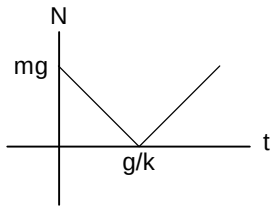
2. A block of mass m is resting inside a smooth horizontal tube. If the tube accelerates vertically downward with acceleration a = kt where k is a positive constant, then correct plot showing the variation of normal reaction received by the block as a function of time (considering that upward direction of normal reaction is positive and taken towards positive direction of y axis) will be



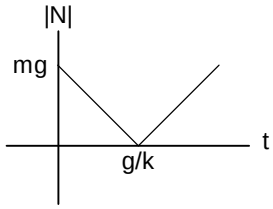
(A)



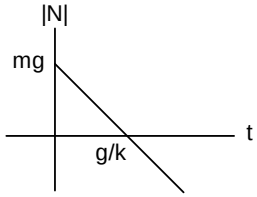
(B)



(C)

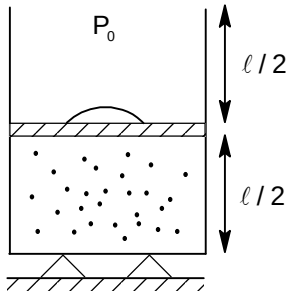


(D)



Space for Rough work

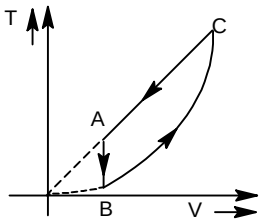
3. Figure shows initial state of an ideal gas trapped in a container with conducting walls and a piston (mass m) which can move without any friction. The container is placed on point supports and its walls are conducting. Assuming that atmospheric pressure is P_0 and the mass of the ideal gas is negligible as compared to the mass of the piston and the mass of container. Take the cross section area of piston to be A. The piston is slowly lifted by an external agent and held in its position. Let M be the maximum mass of container so that it may "lift off" while pulling the piston upwards and P_i be the pressure of ideal gas in initial state. Pick the correct choice:



(A) $P_i = P_0 + \frac{mg}{A}$ (B) $P_i = \frac{P_0}{2} + \frac{mg}{2A}$

(C) $M = \frac{m}{2}$ (D) $M = \frac{P_0 A}{2g} - \frac{m}{2}$

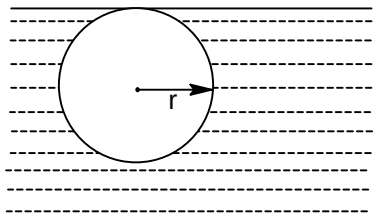
4. A monoatomic gas undergoes a cyclic process as shown in the figure. The ideal gas follows the equation $T = bV^2$ for the process B $\rightarrow$ C. During the cycle; the ratio of maximum to minimum pressure is 2 and heat supplied to the gas in B $\rightarrow$ C is $Q_{B \rightarrow C}$ = 120J. Assuming usual meanings for the symbols, pick correct option(s):—



(A) $P_A = P_C$ (B) $\frac{T_A}{T_B} = 2$

(C) $\frac{T_C}{T_B} = 2$ (D) $Q_{C \rightarrow A \rightarrow B}$ = –130J.

5. Figure shows a solid sphere positioned just below the surface of a fluid and held there. Let the density of the fluid be σ, force applied by fluid on the upper half of the sphere be F_1 and the force on lower half be F_2 (excluding the contribution due to atmospheric pressure). Pick correct option(s):-

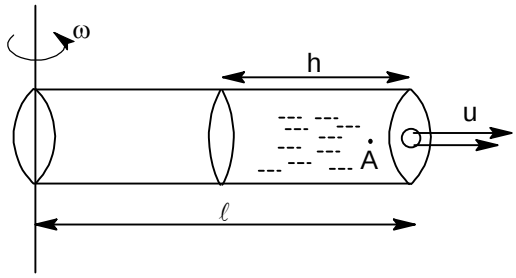


(A) $F_1 = \frac{1}{3}\pi r^3 \sigma g$ (B) $F_2 = \frac{5}{3}\pi r^3 \sigma g$

(C) $F_1 = \frac{2}{3}\pi r^3 \sigma g$ (D) $F_2 = \frac{6}{3}\pi r^3 \sigma g$

Space for Rough work

6. Figure shows a thin tube open at one end rotating with constant angular speed ω. The other end of the tube has a small opening. Tube contains a fluid of density ρ which effuses with a speed u relative to the tube. Let p_A be the pressure at the point in the tube just behind the opening. Assume the atmospheric pressure to be p_0 and pick correct option(s) for the instant when $h = \frac{\ell}{2}$.



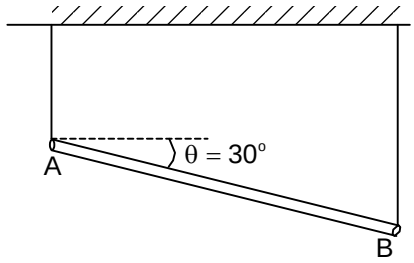
(A) $p_A = p_0 + \frac{\rho\omega^2\ell^2}{4}$

(B) $p_A = p_0 + \frac{3}{8}\rho\omega^2\ell^2$

(C) $u = \frac{\omega\ell}{2}$

(D) $u = \frac{\sqrt{3}}{2}\omega\ell$

7. Figure shows a thin rod (mass M, length L) suspended in a vertical plane by tying ideal strings at ends A and B. Initially, rod makes an angle 30° with horizontal and is at rest. Now; the string connected to the end B is cut. Let T be the tension in the string connected to A and a_A be the magnitude of acceleration of point A just after the "cutting" event. Pick correct option(s):—



(A) $T = \frac{3mg}{13}$

(B) $a_A = \frac{3g}{13}$

(C) $T = \frac{4mg}{13}$

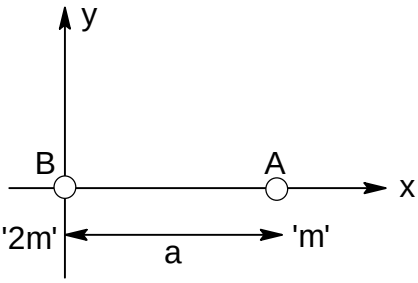
(D) $a_A = \frac{3\sqrt{3}g}{13}$

Space for Rough work

8. Figure shows the initial position of a system of two particles. Given that centre of mass of the system remains at rest and particle A moves in a trajectory given by

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Pick the correct option(s) for the nature of B's trajectory and the coordinates of B at the instant when A was at the position (0, b).



(A) Trajectory of B is elliptical

(B) Trajectory of B is circular

(C) $\left(0,\frac{-b}{2}\right)$

(D) $\left(\frac{a}{2},\frac{-b}{2}\right)$

9. A and B are two hydrogen like atoms such that $m_B = 2m_A$. Also; the number of protons and neutrons in the two nuclei are equal. Given that difference of photon energy corresponding to the first Balmer lines emitted by A and B is 5.667 eV. Let Z_A and Z_B be the atomic numbers of A and B respectively.

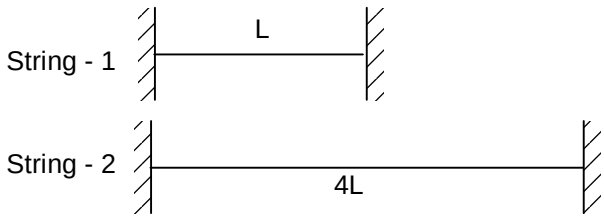
(A) $Z_A = 4$

(B) $Z_A = 1$

(C) $Z_B = 2$

(D) $Z_B = 8$

10. Two strings made of same material, same cross-section but different lengths are fixed at both ends and excited in their respective fundamental modes such that total oscillation energy is the same in both cases. let f_1, f_2 be the oscillation frequencies and A_1, A_2 be the amplitudes of oscillation of the antinodes in the two cases. Assuming both strings to be under same tension, pick the correct option(s).



(A) $\frac{f_1}{f_2}=\frac{1}{4}$

(B) $\frac{A_1}{A_2}=\frac{1}{2}$

(C) $\frac{f_1}{f_2}=\frac{4}{1}$

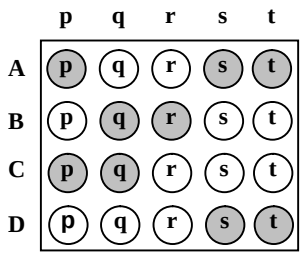
(D) $\frac{A_1}{A_2}=\frac{2}{1}$

Space for Rough work

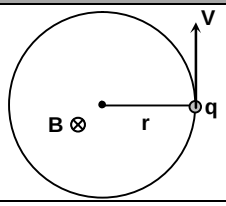
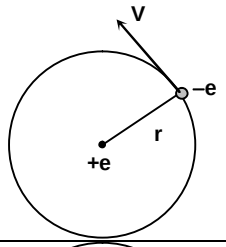
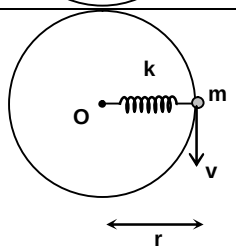
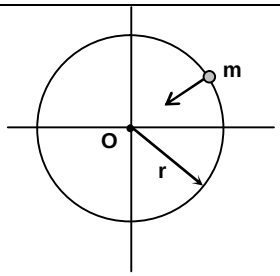
SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

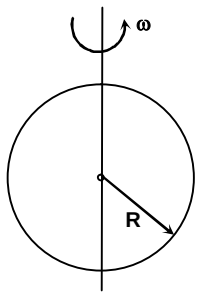


11. Assume Bohr's law of quantization to be true. Column-I lists four situations where a particle is revolving in a circular orbit. Column-II lists the dependence of orbital radius and orbital speed on the order of orbit 'n'.

Column –I			Column -II	
(A)		Charge particle (q, m) revolving in a uniform magnetic field B.	(p)	$r \propto n^2$
(B)		An electron revolving around a hydrogen nucleus with constant speed.	(q)	$v \propto \sqrt{n}$
(C)		A particle connected to a hypothetical spring (with zero natural length) and revolving on a smooth surface.	(r)	$v \propto \frac{1}{n}$
(D)		A small particle revolving about origin with constant speed under a central force given by $F = \alpha\frac{v}{r}$ (α a constant)	(s)	$r \propto \sqrt{n}$
			(t)	$r \propto n$

Space for Rough work

12. Figure shows a solid sphere (M, R) rotating with an angular speed ω. Coefficient of linear expansion is $\alpha = 10^{-5}/°C$. The temperature is increased by 10°C. Column-I lists four quantities and column –II lists the corresponding values/nature of changes in these quantities.



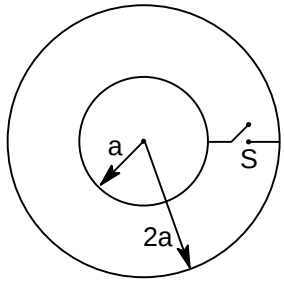
Column – I		Column - II	
(A)	Kinetic energy.	(p)	0%
(B)	Angular speed	(q)	Increases
(C)	Angular momentum	(r)	Decreases
(D)	Moment of inertia	(s)	0.02%
		(t)	0.01%

Space for Rough work

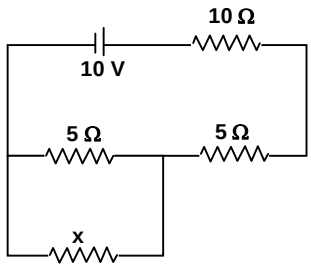
SECTION – C
(Single digit integer type)

This section contains **8** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

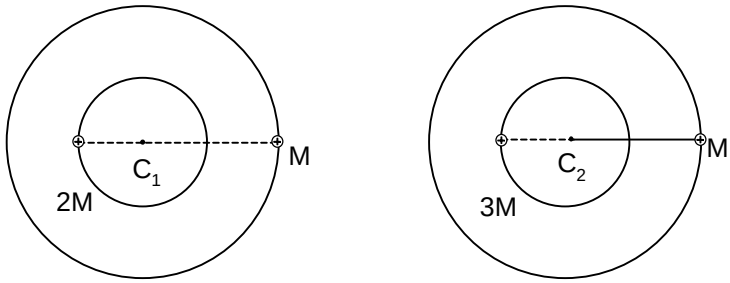
13. Figure shows two concentric conducting spherical shells, each initially carrying a charge Q. The amount of heat generated in the system when the two shells are joined by a conducting wire by closing the switch S is $\frac{(Q/n)^2}{\pi\varepsilon_0 a}$ where n is an integer. Calculate the value of n.



14. Figure shows a battery circuit with an unknown resistance x. If the value of x for maximum power dissipation in unknown resistance is $\frac{5}{n}\Omega$ (n is an integer); find n.



15. Figure shows two binary star systems such that the distance of lighter star from the centre of rotation is same in both cases. If the ratio of time periods of rotation is $\frac{T_1}{T_2}=\frac{n}{8}\sqrt{\frac{3}{2}}$, where n is an integer, find n.



16. Consider the following nuclear reaction

$${}^2_1D + {}^2_1D \longrightarrow {}^3_1T + {}^1_1P$$

Given that $m({}^2_1D) = 2.01458$ amu,

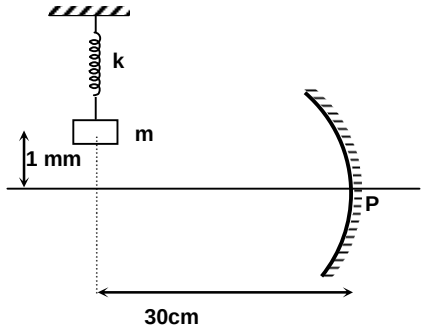
$m({}^3_1T) = 3.01605$ amu,

$m({}^1_1p) = 1.00728$ amu.

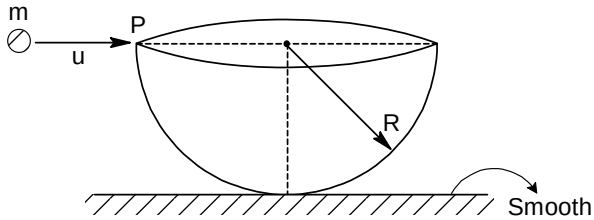
The mass of deuterium $({}^2_1D)$ required per day in order to produce a power output of 10^9W (with 50% efficiency) is $n\cdot 3$ kg where n is a digit. Find n.

Space for Rough work

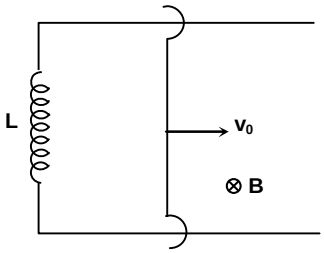
17. Figure shows a small block suspended vertically using an ideal spring. Initially spring is in natural length and the mass is 1 mm above the principal axis of a concave mirror whose focal length is 20 cm. Given that mass m = 10 grams, k = 100 N/m, g = 10 m/s^2; the amplitude of the image of the block formed by concave mirror is A millimetres, find the integer value 'A'.



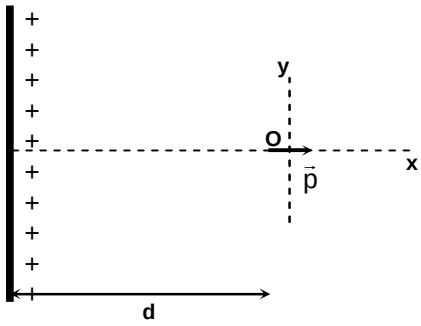
18. A hollow hemispherical bowl of mass m, radius R is placed on a smooth horizontal surface. A particle (also of mass m) strikes the hemi spherical bowl at point P and sticks to it. The velocity of particle just after the collision in vertically upwards direction is $\frac{ku}{25}$; where k is an integer. Find k.



19. A conducting rod of mass m = 0.3 kg and length ℓ = 4m can slide without friction on two parallel conducting rails. The conducting rails are connected via an inductance L = 3mH. This system is placed in a region containing uniform magnetic field B = 1T pointing into the plane. If the rod is given an initial velocity v_0 =2 m/s, it oscillates with an amplitude A cm. Find the value of 2A.



20. An infinite non-conducting plane with uniform charge density σ is kept parallel to yz plane and at a distance 'd' from a dipole $\vec{p}$ which itself is located at the origin. An equipotential surface for this system is spherical, centred at origin, having radius R(<d). Given that $R = \left(\frac{p}{n\pi\sigma}\right)^{1/3}$; find the integer n.



Space for Rough work

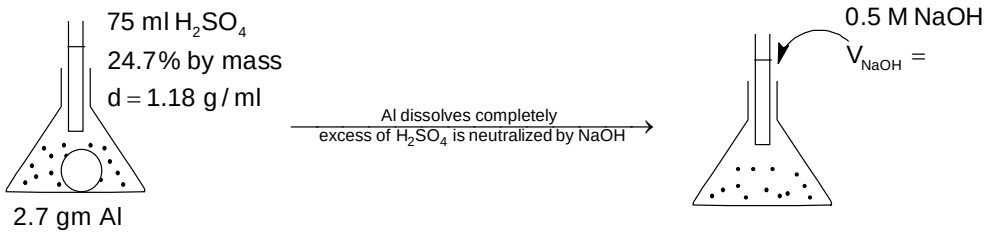
Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **10** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

21. Value of V_{NaOH} is



(A) 291 ml (B) 145.5 ml
(C) 75 ml (D) 582 ml

22. According to Charle's law:

(A) $V \propto \frac{1}{T}$ (B) $\left(\frac{dV}{dT}\right)_P = K$

(C) $\left(\frac{dT}{dV}\right)_P = K$ (D) $\left(\frac{1}{T} - \frac{V}{T^2}\right)_P = 0$

23. FeO crystal has a simple cubic structure and each edge of the unit cell is 5 $\overset{o}{A}$. Taking density of crystal as 4 gm/cc the number of Fe^{2+} and O^{2-} ions present in each unit cell are:

(A) $4Fe^{2+}$ and $4O^{2-}$ (B) $6Fe^{2+}$ and $6O^{2-}$
(C) $2Fe^{2+}$ and $2O^{2-}$ (D) $1Fe^{2+}$ and $1O^{2-}$

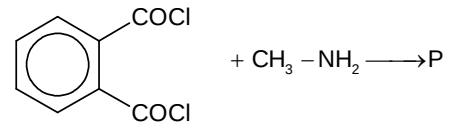
24. Photo electric emission is observed from a surface for frequencies ν_1 and ν_2 of the incident radiation $(\nu_1 > \nu_2)$. If the maximum kinetic energies of the photo electrons in the two cases are in the ratio of 1 : K then the threshold frequency ν_o is given by

(A) $\frac{\nu_2 - \nu_1}{K-1}$ (B) $\frac{K\nu_1 - \nu_2}{K-1}$

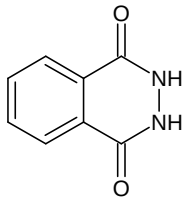
(C) $\frac{K\nu_2 - \nu_1}{K-1}$ (D) $\frac{\nu_2 - \nu_1}{K}$

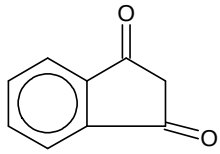
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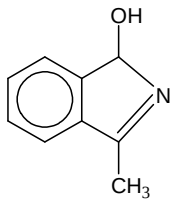
25. Select the correct statement(s)
(A) An adiabatic system can exchange energy with its surroundings.
(B) A thermodynamic property which is intensive is additive.
(C) Work done in free expansion is zero.
(D) For an isolated system, the entropy can never decrease.

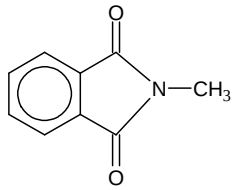
26. 

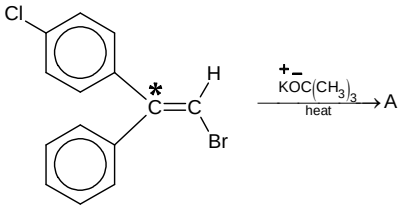
P is?

(A) 

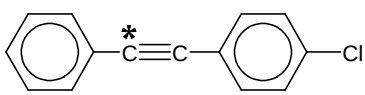
(B) 

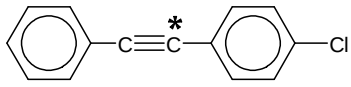
(C) 

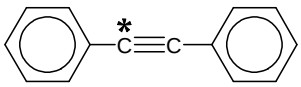
(D) 

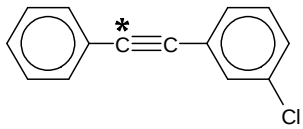
27. 

A is?

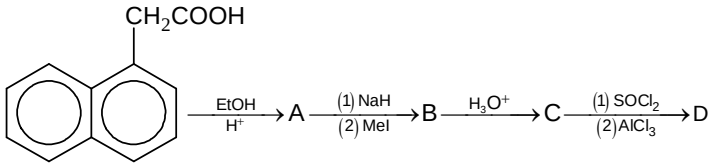
(A) 

(B) 

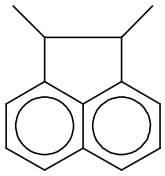
(C) 

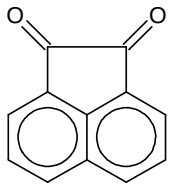
(D) 

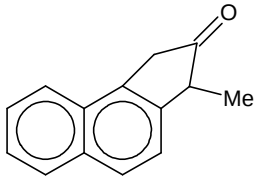
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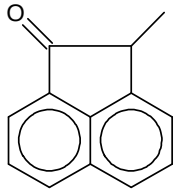
28. 

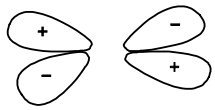
D is?

(A) 

(B) 

(C) 

(D) 

29. Choose the correct statement(s) for O_2 molecule.
As per molecular orbital theory both the unpaired electrons of O_2 molecule are:
(A) in degenerated molecular orbitals
(B) in molecular orbitals which have given shape 

(C) in molecular orbitals which are perpendicular to each other.
(D) all are incorrect.

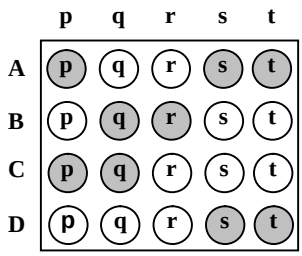
30. Electron affinity of the elements or ions is correctly shown in
(A) $S > O^-$ (B) $O > S^-$
(C) $O^- > S^-$ (D) $N^- > S$

Space for Rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



31. Match the following Column, Column I with Column II:

Column – I		Column – II	
(A)	$\underline{N}(SiH_3)_3$	(p)	$p\pi - d\pi$ back bonding
(B)	$\underline{N}(CH_3)_3$	(q)	sp^3 – hybridization for underlined atom
(C)	$\underline{B}_2H_6$	(r)	$p\pi - p\pi$ back bonding
(D)	$\underline{B}F_3$	(s)	Neither $p\pi - p\pi$ nor $p\pi - d\pi$ back bonding
		(t)	Underlined atom combines with Lewis base

32. Match the following Column, Column I with Column II:

Column – I		Column – II	
(A)	$[Ma_3b_2c]$	(p)	All stereoisomers are optically inactive
(B)	$[Ma_3b_3]$	(q)	Number of geometrical isomers = 2
(C)	$[Ma_3bcd]$	(r)	Number of geometrical isomers = 4
(D)	$[Ma_4bc]$	(s)	Total 3 stereoisomers
		(t)	Only one enantiomeric pair is possible

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **8** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

33. 60 ml of mixture of equal volumes of Cl_2 and an oxide of chlorine, i.e. Cl_2O_n was heated and then cooled back to the original temperature. The resulting gas mixture was found to have volume of 75 ml. On treatment with KOH solution, the volume contracted to 15 ml. Assume that all measurements are made at the same temperature and pressure. Deduce the value of n in Cl_2O_n. The oxide of Cl_2 on heating decomposes quantitatively to O_2 and Cl_2.

34. A hydrogen-like atom (atomic number z) is in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successively emitting two photons of energies 10.2 eV and 17.0 eV respectively. Alternatively, the atom from the same excited state can make a transition to the second excited state by successively emitting 2 photons of energy 4.25 eV and 5.95 eV respectively. Determine the value of (n + z).

35. By passing a certain amount of charge through NaCl solution, 9.2 lit. of Cl_2 were liberated at STP. When the same charge is passed through a nitrate solution of metal M, i.e. $M(NO_3)x$, 7.467 gm of the metal was deposited. If the specific heat of metal is 0.216 cal/gm, what is the value of x (x is integer).

36. A solution containing 28 gm phosphorus (P_n) in 315 gm CS_2 (boiling point 46.3°C) boils at 47.98°C. K_b' for CS_2 is 2.34 K mol^{-1} kg. Calculate value of n. Assume its complete association.

Space for Rough work

37. Find number of species which produce elemental sulphur when treated with H_2S:
$Fe^{3+}, Fe^{2+}, SO_2, H_2O_2, O_3, HNO_3$

38. Out of Be, Mg, Ca, Sr and Ba, total number of metals whose nitrates are decomposed according to following reaction
$2M(NO_3)_2 \xrightarrow{\Delta} 2MO + 4NO_2 + O_2$

39. $\underset{(\text{All isomers})}{C_2FClBrI} \xrightarrow{H_2/Ni} X(\text{excluding stereoisomers})$

$\underset{(\text{All isomers})}{C_4H_8\,(\text{alkene})} \xrightarrow{H_2/Ni} Y(\text{excluding stereoisomers})$

Find the value of X + Y.

40. How many isomers 'X' of C_8H_{10} give only aromatic dicarboxylic acid when react with hot alkaline $KMnO_4$.
How many isomers 'Y' of C_4H_8 give CO_2 when reacts with hot alkaline $KMnO_4$. Find the value of X + Y?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **10** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Let $A_1, A_2 ..... A_7$ be a polygon and $a_1, a_2 ..... a_7$ be the complex numbers representing vertices $A_1, A_2 ..... A_7$. If, $|a_1| = |a_2| = ... |a_7| = R$, then $\sum_{1\le i<j\le 7} |a_i + a_j|^2$

(A) greater than $30R^2$
(B) has minimum value as $35R^2$
(C) has its minimum value in $(25R^2, 45R^2)$
(D) is less than $45R^2$

42. Let $A_1, A_2 ..... A_{26}$ be a regular polygon with 26 sides inscribed in a circle of radius R. Now, A_1', A_7', A_9' be the projections of the orthocentre H of $\Delta A_1A_7A_9$ onto sides A_7A_9, A_1A_9, A_1A_7 respectively. Then

(A) $HA_1' - HA_7' + HA_9' = \frac{2R}{3}$
(B) $HA_7' = 2R\cos A_1 \cos A_9$
(C) $HA_9' = -2R\cos A_1 \cos A_7$
(D) $HA_1' - HA_7' + HA_9' = \frac{R}{2}$

43. Consider f: [0, 1] $\to$ R has a continuous derivative and $\int_0^1 f(x)dx = 0$, then for every $\alpha \in (0, 1)$

$\left|\int_0^1 f(x)dx\right| \le k \cdot \max_{0\le x\le 1}|f'(x)|$ then

(A) $\left[\frac{1}{k}\right] > 5$
(B) $\frac{1}{8}\left[\frac{1}{k}\right] \in [0,1]$
(C) $k \in \left(0, \frac{1}{5}\right)$
(D) $\frac{1}{2k} + 1$ is an odd number

44. Consider the lines, $L_1 : \frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $L_2 : \frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$, then the line along shortest distance can be, constituted by the line of intersection of planes

(A) $4x + y - 5z = 0$
(B) $x - 3y + 5z = 0$
(C) $5x - 7y + 2z = -1$
(D) $7x + y - 8z = 31$

Space for rough work

45. $\vec{p}, \vec{q}, \vec{r}$ be vectors such that $\vec{q} \cdot \vec{r} = 0$ and $\vec{p} \cdot \vec{q} \neq 0$. Let α is real constant such that $\vec{x} \cdot \vec{p} = \alpha$; $\vec{x} \times \vec{q} = \vec{r}$, then $\vec{x} = \lambda_1 \vec{q} + \lambda_2 (\vec{p} \times \vec{r})$ where

(A) $\lambda_1 = \dfrac{\alpha}{\vec{p} \cdot \vec{q}}$ (B) $\lambda_2 = \dfrac{1}{\vec{p} \cdot \vec{q}}$

(C) $\lambda_2 = \dfrac{1}{\vec{r} \cdot \vec{q}}$ (D) $\lambda_1 = \dfrac{\alpha}{\vec{r} \cdot \vec{q}}$

46. Consider a hyperbola $\dfrac{x^2}{36} - \dfrac{y^2}{25} = 1$. A chord QPL meets an asymptote in L and a tangent from L is drawn touching at R. If PM, RE, QN be drawn parallel to the asymptote to meet the other asymptote
PM + QN = λ·RE where

(A) λ is less than 1 (B) λ is less than $\dfrac{5}{2}$

(C) [λ] is 2 (D) [λ] is zero

47. If 4 – horses participate in race. The number of ways in which they can go through finish line is α (Dead heat is possible), then

(A) $\left[\dfrac{\alpha}{6}\right] = 12$ (B) $\left[\dfrac{\alpha+1}{10}\right] = 7$

(C) $\left[\dfrac{100}{\alpha}\right] = 2$ (D) $\dfrac{\alpha}{12} = \beta$, $\beta \in I$

48. Consider an ellipse $\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$. A circle passes through a focus and has its centre on y = 0 and touches the ellipse at A and S is focus, then

(A) [|AS|] is less than 7 (B) |AS| < 4

(C) 5 < |AS| < 6 (D) |AS| > 4

49. Consider 18 – objects (distinct) then the probability that we can take odd number of objects in a draw is

(A) $\dfrac{1}{2}$ (B) $\dfrac{1}{4}$

(C) $\dfrac{1}{8}$ (D) $\dfrac{1}{16}$

50. If f(x) is a polynomial such that $(\alpha + 1)f(\alpha) - \alpha = 0 \ \forall \ \alpha \in N \cup \{0\}$, $\alpha \leq n$, then

(A) f(76) is 1 (B) f(21) is $\dfrac{10}{11}$

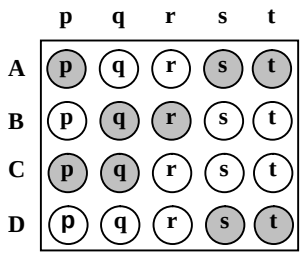
(C) f(37) is $\dfrac{1}{7}$ (D) f(148) is $\dfrac{9}{13}$

Space for rough work

SECTION - B

Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



51. Consider lines as shown L_1 : y = ax, z = c ; L_2 : y = –ax, z = –c ; L_3 : y = z, ax = –c. Then L is a variable line which intersects the given lines ($a \neq b \neq c$), match the following Column-I with Column-II

Column – I		Column – II	
(A)	Locus of L passes through point P where P is	(p)	$(a^2, 1, -c^2)$
(B)	Locus of L is $k_1x^2 - y^2 + k_2z^2 + k_3 = 0$, then (k_1, k_2, k_3) is	(q)	(0, 0, c)
(C)	Locus L intersects the xz-plane in locus L′ which is an ellipse with major axis α, minor axis β, centre as (α_1, β_1) where $(\alpha, \beta, \alpha_1 + \beta_1)$ is $(a^2 > 1)$	(r)	$\left(\frac{2c}{a}, 1, 0\right)$
(D)	Locus L intersects the xy-plane in hyperbola with transverse axis α', conjugate is β', centre as $\left(\alpha_1', \beta_1'\right)$ where $\left(\alpha', \beta', \alpha_1', \beta_1'\right)$ is $\left(\frac{c^2}{a^2} > 1\right)$	(s)	$\left(2c, \frac{2c}{a}, 0\right)$
		(t)	$\left(\frac{c}{a}, 0, 0\right)$

Space for rough work

52. Consider a ΔABC and points K, L, M on the sides AB, BC, CA respectively such that $\frac{AK}{AB}=\frac{BL}{BC}=\frac{CM}{CA}=\frac{1}{3}$, match the following Column-I with Column-II

Column – I		Column – II	
(A)	$\frac{\text{Area}(\Delta AKM)}{\text{Area}(\Delta ABC)}$ is	(p)	$\frac{1}{3}$
(B)	$\frac{\text{Area}(\Delta KML)}{\text{Area}(\Delta ABC)}$ is	(q)	$\frac{1}{\sqrt{3}}$
(C)	$\frac{\text{Radius of incircle of } \Delta BAC}{\text{Radius of circumcircle of } \Delta ABC}$ is	(r)	3
(D)	$\angle ABC=\frac{\pi}{K}$	(s)	$\frac{2}{9}$
		(t)	greater than 1

Space for rough work

SECTION – C
(Single digit integer type)

This section contains **8** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

53. If A is 5×4 matrix and B is 4×5 matrix, then $2(|AB - I_5| + |BA - I_4|) + 1$ is _____

54. Consider a straight line $\frac{x}{a}+\frac{y}{b}=1$, such that it cuts the asymptotes of hyperbola xy = 1 in points A and B & the hyperbola itself in P and Q, then $\frac{AP}{BQ}=\lambda$ where $2\lambda + 1$ is _____

55. If $\alpha, \beta, \gamma \in R^+$ such that $\alpha\gamma=\frac{1}{\beta}$, then maximum value of $\alpha^{\beta+\gamma}\cdot\beta^{\alpha+\gamma}\cdot\gamma^{\alpha+\beta}$ is _____

56. If $a_1 = a$, $a_2 = b$, $a_{n+2} = \frac{a_{n+1}+a_n}{2}$, $n \geq 0$ then, $\lim\limits_{n\to\infty} a_n$ is $\frac{a+\lambda_1 b}{\lambda_2}$; $(\lambda_1, \lambda_2 \in I)$ where $\lambda_1 + \lambda_2$ is _____

57. Let z_1, z_2, z_3 be complex numbers (not all real) such that $|z_1| = |z_2| = |z_3| = 1$ and $2(z_1 + z_2 + z_3) - 3z_1z_2z_3$ is real. Then, Max(arg(z_1), arg(z_2), arg(z_3)) (Given that argument of z_1, z_2, z_3 is positive) has minimum value as $\frac{k\pi}{6}$ where (k + 2) is _____

58. A bag contains 2017 red balls and 2017 black balls. We remove two balls at a time repeatedly and (1) discard them if they are of same colour (2) discard the black ball and return to the bag the red ball if they are of different colours. Then the probability that this process will terminate with one red ball is 'p' where 3p is _____

59. If $I=\int_0^{\pi}\ln\left(1-2a\cos x+a^2\right)dx$ where $I = \lambda \ln a$, then $[\lambda]$ is _____

60. Consider a function f: [0, 1] $\to$ R satisfying $\int_0^1\left(f(x)\left(x-f(x)\right)\right)dx=\frac{1}{12}$ then, f′(1) is k where $\left[\frac{1}{k}\right]$ is _____

Space for rough work